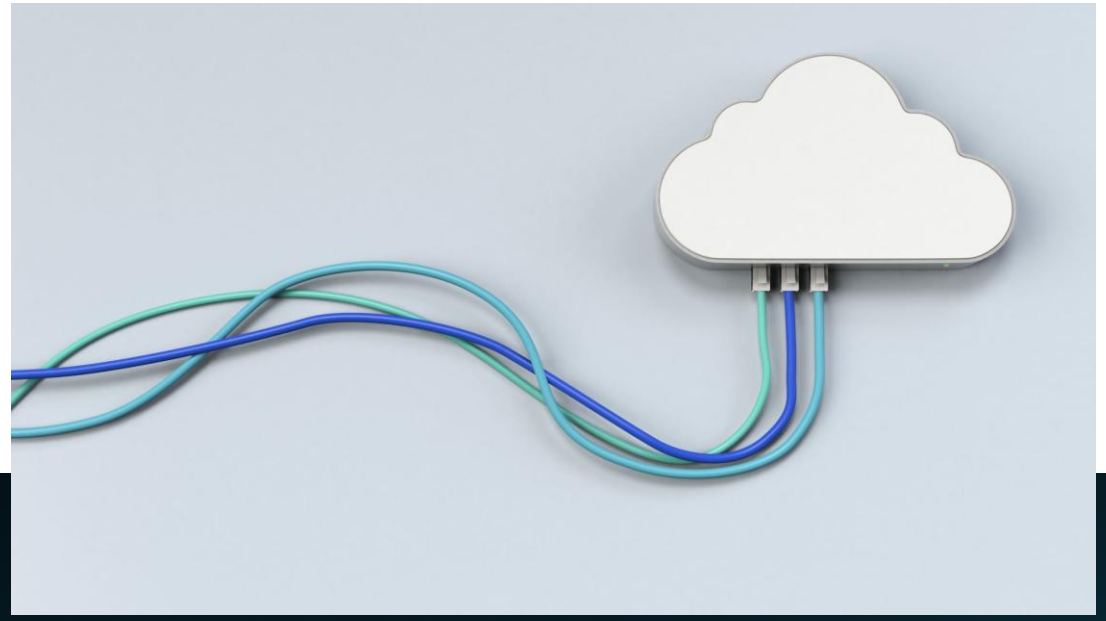
AWS Cloud Infrastructure Deployment



Students ID:

B00173541 Sofiia Prymak

B00172387 Livia Mathew

B00168689 Róisín Aylward



High-Level Problems

Main Problems Identified:

- No structured security framework
- No visibility into threats
- Weak access control practices
- No formal backup or recovery plan
- Outdated or inconsistent security policies

Security Gaps

Governance Gaps

- No defined security roles
- No documented policies

Technical Gaps

- No MFA
- No network segmentation
- No centralized logging
- No automated updates

Operational Gaps

- No incident response plan
- No disaster recovery strategy
- No employee security training



Key Security Risks

1. Unauthorized Access Risk

- Weak authentication → attackers can enter systems.

2. Data Loss Risk

- No backups → business-critical data could be permanently lost.

3. Service Outage Risk

- Single-point-of-failure systems → downtime during failures.

4. Malware / Ransomware Risk

- Unpatched systems → high vulnerability to attacks.

5. Insider Threat Risk

- No monitoring → malicious or accidental actions go unnoticed.

SECURITY MANAGEMENT & RISK AWARENESS



- Identifying key assets such as data, and systems.
- Recognising the potential threats.
- Understanding the weaknesses in the system.
- The clear roles & responsibilities.
- The lessons from past attacks.

LIVIA MATHEW - B00172387

Protecting the System & Detecting Threats





- Limiting access to sensitive data.
- Separating public, and private systems.

- **Controlling user access.**
- **Monitoring the system activity.**
- **Detecting threats only, and earlier.**

- The clear incident response plan in place.
- Fast reactions to the threats.
- The backup systems at hand.
- Restoring services quickly.
- Maintaining customer trust.

Responding to Incidents/ Threats & Recovery



Secure, Scalable & Reliable Solution

- Stronger system security.
- The reduced business risk.
- The improved reliability.
- Supporting business growth.
- Protecting customer trust.



Recommended Controls & Strategies

Róisín Aylward

GOVERN	• Develop an Information Security Policy. • Train staff on policies. • Ensure separation of duties across all roles.
IDENTIFY	• Conduct quarterly risk assessments. • Create an asset inventory. • Classify all Personally Identifiable Information.
PROTECT	• Enforce principle of least privilege. • Encrypt data at rest & in transit. • Apply strict firewall rules.

Recommended Controls & Strategies

Róisín Aylward

DETECT

- Enable logging of user activity.
- Set alarms for unusual logins and traffic.
- Monitor database access patterns.

RESPOND

- Document an Incident Response Plan.
- Define containment steps.
- Establish GDPR breach notification workflows.

RECOVER

- Document a Business Continuity Plan.
- Automate encrypted backups.
- Test recovery procedures regularly.

Conclusion

Róisín Aylward

1. WinLocal's ransomware attack exposed critical gaps. AWS cloud migration will embed security by design.
2. NIST CSF 2.0 gives us a clear six function roadmap.
3. Staff training and policies must be implemented alongside the infrastructure.
4. Risk assessments and incident response plans will minimise the impact of future events.



Thank you for listening! We hope you have a good day!